LECTURE 5

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In this talk, we will prove the main theorems in the Section 5 and Section 6 of [BSY22]. We first prove that for any $T(n)$-local algebra R there always exists a perfect algebra A of Krull dimension 0 and a nilpotence-detecting map $R \to E(A)$. As a consequence, we can map any $T(n)$-local algebra to the Lubin-Tate theory of some algebraic closed field L.

In addition, we define the Nullstellensatzian objects in a presentable category. We show the Nullstellensatzian theorem. That is, the Nullstellensatzian objects in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ are exactly the Lubin-Tate theories attached to algebraic closed fields. At the end of this talk, we will build an analogy of Zariski spectra on $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$.

1. Recollection

We briefly recollect some key results in the previous talks. Recall that we introduced the Lubin-Tate functor. More explicitly, we have:

Proposition 1.1. *Let k be a perfect field of character p and we fix a formal group $\mathbb{G}$ of height n. There is an adjunction*

$$E(-) : \mathrm{Perf}_k \rightleftharpoons \mathrm{CAlg}^{\wedge}_{E(k)} : (-)^{\flat}$$

which we refer to as the Lubin-Tate-tilt adjunction, such that:

1. *$E(-)$ is fully faithful.*
2. *The essential image of $E(-)$ consists of the rings R such that R is even periodic, $\pi_0(R)/\mathfrak{m}$ is perfect and $R//\mathfrak{m}$ is even.*

We have shown that the Lubin-Tate theory is cofree. That is, there is an adjunction

$$(U_{\mathbb{T}}(-)/\mathfrak{m})^{\flat} : \mathrm{Alg}_{\mathbb{T}} \rightleftharpoons \mathrm{Perf}_k : \pi_0 E(-)$$

such that $\pi_0 E(-)$ becomes a fully faithful right adjoint.

We proved the cofreeness by showing that the evaluation maps

$$\overline{ev}_A : \pi_0 E(A) \to W_{\mathbb{T}}(\pi_0 E(A)/\mathfrak{m})$$

become isomorphisms. As a result, we have

Corollary 1.2. *Suppose we have a map $f : \pi_0 E(A) \to \pi_0 R$. Then f is injective if and only if $f/\mathfrak{m}$ is injective.*

Proof. We can approximate f by the evaluation map. After applying $\overline{ev}_A$, we obtain a commutative diagram

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$$\begin{array}{ccc} \pi_0 E(A) & \xrightarrow{\overline{ev}_A} & W_{\mathbb{T}}(\pi_0 E(A)/\mathfrak{m}) \\ {\scriptstyle f}\downarrow & & \downarrow \\ \pi_0 R & \longrightarrow & W_{\mathbb{T}}(\pi_0 R/\mathfrak{m}) \end{array}$$

The above arrow is an isomorphism and the right arrow is an injection, which forces f to be an injection. Here we use the fact that $W_{\mathbb{T}}(-)$ preserves injections. □

Throughout the last talk, we introduced the nilpotence detecting map. We have the following notation.

Definition 1.3. Let $f : C \to D$ be a morphism in $\mathcal{C}$ and R be an object. We say that $f : C \to D$ has right lifting property with respect to R and denote $f \perp R$ if we always have a lifting

$$\begin{array}{ccc} C & \longrightarrow & R \\ {\scriptstyle f}\downarrow & \nearrow & \\ D & & \end{array}$$

By Quillen's small object argument, we can show the following.

Proposition 1.4. *Let $S = \{f_s : C_s \to D_s\}$ be a family in $\mathcal{C}$ consisting of nilpotence-detecting morphisms. For any $R \in \mathcal{C}$, there exists an extention $R \to R'$ such that*

(1) $R \to R'$ detects nilpotence.
(2) For any $f_s \in S$, $f_s \perp R'$.

The spirit of the nilpotence detection is that we can convert the properties of objects to the properties of morphisms. We have a correspondence

$$\{\text{properties of the objects}\} \Leftrightarrow \{\text{family of morphisms}\}$$

such that an object R satisfies these properties if and only if the family of morphisms has right lifting property with respect to R. For example, we can use this principle to detect rings of Krull dimension 0.

Proposition 1.5. *Let $R \in \mathrm{CRing}$. R is of Krull dimension 0 if and only if $f \perp R$, where f is defined as*

$$f : \mathbb{Z}[t] \xrightarrow{t \mapsto (t,0)} \mathbb{Z}[t^{\pm 1}] \times \mathbb{Z}$$

Variant 1.6. We have the followings.

(1) Let $A \in \mathrm{Perf}_k$ be a perfect $\mathbb{F}_p$ algebra. A is of Krull dimension 0 if and only if $f' \perp R$, where f' is defined as

$$f' : \mathbb{F}_p[t] \xrightarrow{t \mapsto (t,0)} \mathbb{F}_p[t^{\pm 1}] \times \mathbb{F}_p$$

(2) Let $\mathcal{C} \in \mathrm{CMon}(\mathrm{Pr}_{st})$ be a p-complete presentable symmetric monoidal stable category. We have a Witt-tilt adjunction

$$\mathbb{W}_{\mathcal{C}}(-) : \mathrm{Perf} \rightleftharpoons \mathrm{CAlg}(\mathcal{C}) : (-)^\flat$$

$R^\flat$ is of Krull dimension 0 if and only if $f'' \perp R$, where f'' is defined as

$$f' : \mathbb{1}_{\mathcal{C}}[t] \xrightarrow{t \mapsto (t,0)} \mathbb{1}_{\mathcal{C}}[t^{\pm 1}] \times \mathbb{1}_{\mathcal{C}}$$

We will see that the above spirit is the main ingredient for the proof of the first theorem.

2. Proof of Theorem D

We first prove the Theorem D of [BSY22].

Theorem 2.1. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. Then there exists a nilpotence-detecting morphism $R \to E(A)$ such that A is reduced and of Krull dimension 0.*

Once we obtain this result, the following is immediate.

Corollary 2.2 ([BSY22]; Theorem D)**.** *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. Then there exists a morphism $R \to E(L)$, where L is an algebraic closed field.*

We begin with the simplified case of height 0.

2.1. **The case of height 0.** Recall that when the height equals 0, everything is of character 0 and "perfect" means "reduced". In this case, $\pi_* E(A) = A[u^{\pm 1}]$ where u is a generator in degree 2.

Note that $\mathbb{Q} \to E(\mathbb{Q})$ admits a retraction. Therefore, the base change functor $- \otimes_{\mathbb{Q}} E(\mathbb{Q})$ is conservative, which implies that $\mathbb{Q} \to E(\mathbb{Q})$ detects nilpotence. As a consequence, it suffices to show

Theorem 2.3. *Let $R \in \mathrm{CAlg}^{\wedge}_{E(\mathbb{Q})}$. Then there exists a nilpotence-detecting morphism $R \to E(A)$ such that A is reduced and of Krull dimension 0.*

The proof idea is to convert the properties of rings to a family of morphisms. We want to find an extension $R \to R'$ such that $R' \simeq E(A)$. Therefore, R' should at least satisfy the following properties:

(1) R' is even.
(2) $\pi_0 R'$ is reduced and is of Krull dimension 0.

The following proposition points to the morphisms that detect these properties.

Proposition 2.4. *We define $f_0 : E(k)[t] \to E(k)[t^{\pm 1}] \times E(k)$ and $h_0 : E(k)\{z^1\} \to E(k)$. Let $S_0 = \{f_0, h_0\}$ be the family of morphisms in $\mathrm{CAlg}_{E(k)}(\mathrm{Sp}_{\mathbb{Q}})$ and let R be an object. Then $S_0 \perp R$ if and only if $\pi_0(R)$ is reduced of Krull dimension 0 and $\pi_*(R) = \pi_0(R)[u^{\pm 1}]$.*

Proof. By definition, it is clear that $h_0 \perp R$ if and only if $\pi_1(R) = 0$. It remains to show that $f_0 \perp R$ if and only $\pi_0(R)$ is reduced of Krull dimension 0. This is because $f_0 \perp R$ if and only if

$$\pi_0(\mathrm{Map}_{\mathrm{CAlg}_{E(k)}(\mathrm{Sp}_{\mathbb{Q}})}(E(k)[t^{\pm 1}] \times E(k), R)) \to \pi_0(\mathrm{Map}_{\mathrm{CAlg}_{E(k)}(\mathrm{Sp}_{\mathbb{Q}})}(E(k)[t], R))$$

is surjective. By freeness of $E(k)[t]$, this is equivalent to say that

$$\mathrm{Map}(\mathbb{Z}[x^{\pm 1}] \times \mathbb{Z}, \pi_0(R)) \to \mathrm{Map}(\mathbb{Z}[x], \pi_0(R)).$$

is surjective. It concludes from Variant 1.6.

□

Proposition 2.5. *The maps $\{f_0, h_0\}$ defined above detect nilpotence.*

Therefore, one can use the small object argument to conclude that one can always extend R to some R', such that R' is even and $\pi_0(R')$ is reduced of Krull dimension 0.

Unfortunately, this does not imply $R' = E(A)$. To complete the proof, we need to map R' to its residue fields. For $x \in \mathrm{Spec}(\pi_0(R'))$, let F_x denote the residue field of $\pi_0(R')$ at x. We can construct ring spectra R'_x such that $\pi_*(R'_x) \cong \pi_0(R'_x)[u^{\pm 1}] \cong F_x[u^{\pm 1}]$. We claim $R'_x \simeq E(F_x)$. Then we have a map

$$R' \to \prod_{x \in \mathrm{Spec}(\pi_0 R')} E(F_x) \simeq E(\prod_{x \in \mathrm{Spec}(\pi_0 R')} F_x)$$

which detects nilpotence since it induces a surjection on $\mathrm{Spec}(\pi_0(-))$. And $\prod_{x \in \mathrm{Spec}(\pi_0 R')} F_x$ is reduced of Krull dimension 0.

To show that $R'_x \simeq E(F_x)$, first note that $E(F_x) \simeq E(k) \otimes_k F_x$. By the ring structure, it suffices to construct a map $F_x \to R'_x$ such that it induces an isomorphism on π_0. This can always be done, since one can consider the Postnikov tower of R'_x and extend $F_x \simeq \tau_{\leqslant 0} R'_x$ along the square-zero extensions.

2.2. **The case of positive height.** From now on, we will focus on positive height. We fix a perfect field k of characteristic p and a formal group $\mathbb{G}$ of height n. We first note that in this case, $\mathbb{S}_{T(n)} \to E(k)$ detects nilpotence by [DHS88]. It suffices to show:

Theorem 2.6. *Let k be a perfect field of characteristic p, and let $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$. Then there exists a perfect k-algebra A of Krull dimension 0 and a nilpotence detecting map $R \to E(A)$.*

The proof idea is similar. Suppose we have an extension $R \to R'$ such that $R' \simeq E(A)$ for some A of Krull dimension 0, we want to associate the properties of R' with a family of morphisms. The following lemma allows us to summarize these properties.

Lemma 2.7. *Let $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$. $R \simeq E(A)$ for some A of Krull dimension 0 if and only if the following properties are satisfied.*

(1) $R^\flat$ is of Krull dimension 0.
(2) The colocalization map $\pi_0 E(R^\flat) \to \pi_0 R$ is surjective.
(3) $\pi_1(R) = 0$.

Proof. The only if part is obvious, so we prove the if part. First, notice that the condition (1) implies that $\pi_0 E(R^\flat) \to \pi_0 R$ is injective. Indeed, by Corollary 1.2, it suffices to check after modulo $\mathfrak{m}$. Therefore, we need to show that the perfection comparison map $(\pi_0(R)/\mathfrak{m})^\flat \simeq R^\flat \to \pi_0(R)/\mathfrak{m}$ is injective. Since everything is of Krull dimension 0, the kernel is generated by idempotents and meanwhile nilpotent, which must vanish.

Therefore, the first and second conditions make sure that the colocalization map is isomorphic on even degrees. The vanishment of odd degrees then guaranties that it is an equivalence. □

Definition 2.8. We define the following three maps in $\mathrm{CAlg}^{\wedge}_{E(k)}$:

(1) $f : E(k[t]) \to E(k[t^{\pm 1}]) \times E(k)$ as in Variant 1.6.
(2) $g : E(k)\{z^0\} \to E(k\{z^0\}^\sharp)$, which is to be defined soon
(3) $h : E(k)\{z^1\} \xrightarrow{z^1 \mapsto 0} E(k)$.

We will prove the following.

Theorem 2.9. *Let f, g, h be defined as above. Then f, g, h detect nilpotence. Moreover, $f \perp R$ if and only if R satisfies (1) in Lemma 2.7; $g \perp R$ if and only if R satisfies (2); $h \perp R$ if and only if R satisfies (3).*

Once we prove this theorem, we can use the small object argument to conclude that for any $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$, we can find an extension $R \to R'$ which detects nilpotence, and such that $R' \simeq E(A)$ for some A of Krull dimension 0. We win.

The statements about f come from the previous talk. We need only to discuss the statement for g and h.

2.3. **The map g.** We begin with some notations. We use $k\{z^0\}$ to denote the symmetric polynomial. We have an equivalence $k\{z^0\} \cong \pi_0(E(k)\{z^0\})/\mathfrak{m}$. To define g, we need the following lemma.

Lemma 2.10. *Let A be a perfect k-algebra. Then the functor*

$$(\pi_0(-)/\mathfrak{m})^\sharp : \mathrm{CAlg}^{\wedge}_{E(k)} \to \mathrm{Perf}_k$$

induces an isomorphism

$$\pi_0\mathrm{Map}(E(k)\{z^0\}, E(A)) \to \mathrm{Map}_{\mathrm{Perf}_k}(k\{z^0\}^\sharp, A)$$

Proof. One can factor the map as

$$\begin{aligned}\pi_0\mathrm{Map}(E(k)\{z^0\}, E(A)) &\xrightarrow{\pi_0} \mathrm{Map}_{\mathbb{T}}(\pi_0(E(k)\{z^0\}), \pi_0(E(A))) \\ &\xrightarrow{(U_{\mathbb{T}}(-)/\mathfrak{m})^\sharp} \mathrm{Map}_{\mathrm{Perf}_k}(k\{z^0\}^\sharp, A).\end{aligned}$$

The second arrow is an isomorphism by cofreeness of Lubin-Tate theories. It suffices to show the first isomorphism. Consider the diagram

$$\begin{array}{ccccc}\pi_0\mathrm{Map}(E(k)\{z^0\}, E(A)) & & \xrightarrow{\quad\simeq\quad} & & \pi_0\Omega^\infty E(A) \\ \downarrow{\scriptstyle \pi_0} & & & & \downarrow{\scriptstyle \simeq} \\ \mathrm{Map}_{\mathbb{T}}(\pi_0(E(k)\{z^0\}), \pi_0(E(A))) & \longrightarrow & \mathrm{Map}_{\mathbb{T}}(\mathbb{T}(\pi_0 E(k)), \pi_0(E(A))) & \xrightarrow{\simeq} & \pi_0 E(A)\end{array}$$

where $\mathbb{T}(\pi_0 E(k))$ is the free $\mathbb{T}$-algebra on one generator. The bottom left horizontal map is induced by precomposition with the map $\mathbb{T}(\pi_0 E(k)) \to \pi_0(E(k)\{z^0\})$, which is an isomorphism since $\pi_0 E(A)$ is $\mathfrak{m}$-adically complete and $\mathbb{T}(\pi_0 E(k))^{\wedge}_{\mathfrak{m}} \cong \pi_0(E(k)\{z^0\})$ (see [Rez09, Proposition 4.17]). The top and bottom right horizontal equivalences are because of freeness. □

Definition 2.11. We define $g : E(k)\{z^0\} \to E(k\{z^0\}^\sharp)$ as the map that associates $\mathrm{id} : E(k\{z^0\}^\sharp) \to E(k\{z^0\}^\sharp)$.

Our next aim is to show that g satisfies the desired properties.

Proposition 2.12. *We have the following properties:*

1. *The functor $- \otimes_{E(k)\{z^0\}} E(k\{z^0\}^\sharp)$ induced by g is conservative. In particular, g detects nilpotence.*
2. *For $R \in \mathrm{CAlg}^{\wedge}_{E(k)}$, we have $g \perp R$ if and only if $\pi_0(E(R^\flat)) \to \pi_0(R)$ is surjective.*

Proof. The first assertion is by reducing to $\mathrm{Mod}_{k\{z^0\}}$ and the fact that $k\{z^0\}^\sharp$ is free as a $k\{z^0\}$-module. The second assertion is because we have

$$\begin{aligned}\pi_0\mathrm{Map}(E(k\{z^0\}^\sharp), R) &\cong \pi_0\mathrm{Map}_{\mathrm{Perf}_k}(k\{z^0\}^\sharp, R^\flat)\\ &\cong \pi_0\mathrm{Map}(E(k)\{z^0\}, E(R^\flat))\\ &\cong \pi_0(E(R^\flat))\end{aligned}$$

The first isomorphism is from the Lubin-Tate-tilt adjunction, the second is by the above lemma, and the last is by the definition of $E(k)\{z^0\}$. □

Once we check the properties of f and g, using small object argument, we obtain the following intermediate consequence.

Corollary 2.13. *Let $R \in \mathrm{CAlg}^\wedge_{E(k)}$. Then there exists a nilpotence detecting map $R \to R'$ such that*

$$E(R'^\flat) \to R'$$

induces an isomorphism on π_0 and $\pi_0(R')$ is reduced of Krull dimension 0.

2.4. **The map h.** We next consider the map $h : E(k)\{z^1\} \xrightarrow{z^1\mapsto 0} E(k)$. We need to show that h detects nilpotence. Our strategy is to convert the problem of nilpotence detection to the problem of the existence of special filtrations.

We make some preparations. Let $\alpha \in \pi_1 R$. We use $h_\alpha : R \to R//\alpha$ to denote the $\mathbb{E}_\infty$-quotient of R by α. That is, the commutative algebra map obtained by base-change along h. Recall the definition of phantom maps:

Definition 2.14. Let $\mathcal{C}$ be a rigid category. A map $f : a \to b$ is called phantom if for any $g : c \to a$ with $c \in \mathcal{C}^\omega$, $f \circ g$ is null. f is called $\otimes^m$-phantom if $f^{\otimes m}$ is phantom.

That the fiber of a map is phantom implies the map itself detects nilpotence. We will prove the following.

Proposition 2.15. *Let $R \in \mathrm{CAlg}^\wedge_{E(k)}$ such that $\pi_0(R)$ is reduced and $\alpha \in \pi_1(R)$. Then the fiber of h_α is $\otimes^2$-phantom. In particular, h_α detects nilpotence.*

Once we prove this, it follows that h detects nilpotence. This is because by Corollary 2.13, we can always find a map $q\colon E(k)\{z^1\} \to R$ such that R is reduced. Consider the commutative square

$$\begin{array}{ccc} E(k)\{z^1\} & \xrightarrow{q} & R \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h_\alpha} \\ E(k) & \longrightarrow & R//\alpha \end{array}$$

The top row detects nilpotence by assumption, and the right vertical map detects nilpotence by the proposition. As a consequence, h detects nilpotence as well.

We will convert the problem of $\otimes^2$-phantom to the existence of certain filtrations. We will need the following notation.

Notation 2.16. We only consider the ascending filtration. That is, the filtration indexed by $\mathbb{N}$ such that 0 is the initial object. For a symmetric monoidal stable category $\mathcal{C}$, we have the following usual functors

$$\mathcal{C} \underset{\text{ct}}{\overset{\text{colim}}{\leftarrow}} \text{Fil}(\mathcal{C}) \xrightarrow{\text{gr}} \text{Gr}(\mathcal{C}) \xrightarrow{\oplus} \mathcal{C}$$

Definition 2.17. Let $\mathcal{C}$ be a symmetric monoidal stable ∞-category. For $M \in \text{Gr}(\mathcal{C})$, we say that M is **purely even** if for all i, M_i is a finite direct sum of even suspensions of $\mathbb{1}_{\mathcal{C}}$. In this case, the **rank** of M, which we denote by $\text{rank}(M)$, is the total number of summands in $\bigoplus M_i$. For $M_\bullet \in \text{Fil}(\mathcal{C})$, we say M is purely even of rank r if $\text{gr}(M)$ is purely even of rank r. We say $M_\bullet$ is purely odd of rank r if ΣM is purely even of rank r.

Our aim is to prove the following criterion for $\otimes^2$-phantom maps.

Lemma 2.18. *Let $R \in \text{CAlg}^\wedge_{E(k)}$ such that $\pi_0(R)$ is reduced and let $M_\bullet \in \text{Fil}(\text{Mod}^\wedge_R)$ be purely odd. Then any map $\text{colim}M_\bullet \to R$ is $\otimes^2$-phantom.*

The lemma is based on the following result.

Proposition 2.19. *Let $R \in \text{CAlg}^\wedge_{E(k)}$ such that $\pi_0(R)$ is reduced. Suppose $N_1 \xrightarrow{f} N_2 \xrightarrow{g} \text{ct}(R)$ is a sequence of maps in $\text{Fil}(\text{Mod}^\wedge_R)$ such that N_1 is purely even of finite rank and N_2 is purely odd of finite rank. Then the composite $g \circ f$ is null.*

Proof. First, since N_1 is of finite rank, one can properly choose some $\Sigma^{2j}\text{ct}(R)\langle m\rangle$ such that we have a cofiber sequence

$$\Sigma^{2j}\text{ct}(R)\langle m\rangle \to N_1 \to N_1'$$

where $\text{rank}(N_1') = \text{rank}(N_1) - 1$. By induction along the cofiber sequence, we reduce to the case where $\text{rank}(N_1) = 1$.

When $\text{rank}N_1 = 1$, $g \circ f$ determines a point in $\text{Map}(N_1, \text{ct}(R)) \simeq \text{Map}(\text{colim}N_1, R)$. Since N_1 is of rank 1, this lies in some even homotopy group of R. Note that when R is reduces, $R \to R \otimes \mathbb{Q}$ induces injection on homotopy groups, it therefore suffices to show in the rational case.

We consider the following diagram

$$\begin{array}{ccccc} N_1^{\otimes m} & \xrightarrow{f^{\otimes m}} & N_2^{\otimes m} & \xrightarrow{g^{\otimes m}} & \text{ct}(R)^{\otimes m} \\ \downarrow & & \downarrow & & \downarrow \\ {N_1}_{h\Sigma_m}^{\otimes m} & \longrightarrow & {N_2}_{h\Sigma_m}^{\otimes m} & \longrightarrow & \text{ct}(R)_{h\Sigma_m}^{\otimes m} \end{array}$$

where m is the rank of N_2. In the rational case, one can compute that the outside vertical maps are equivalence and the middle map is zero. Therefore, $(g \circ f)^{\otimes m}$ is null. Now since the even homotopy group of R is reduced, $g \circ f$ is null. □

Proof of Lemma 2.18. The filtered colimit of $\otimes^m$-phantom maps is still $\otimes^m$-phantom. Therefore, we reduce to the case where $M_\bullet$ is of finite rank. Any map $\text{colim}M_\bullet \to R$ is the colimit of $f : M_\bullet \to \text{ct}(R)$, where f can be decomposed as

$$M_\bullet^{\otimes 2} \xrightarrow{f\otimes \text{id}} M_\bullet \xrightarrow{f} \text{ct}(R)$$

which is null by the above lemma and in particular phantom. □

Let $\mathrm{Fib}(h_\alpha) \to R \xrightarrow{h_\alpha} R//\alpha$ be the fiber sequence. To prove Proposition 2.15, it suffices to construct a filtration on this fiber sequence such that it is purely odd on $\mathrm{Fib}(h_\alpha)$. We can first construct a filtration on $R//\alpha$.

Proposition 2.20. *Given a $T(n)$-local commutative algebra R, let $\mathrm{ct}(R)\{z^{1\langle 1\rangle}\}$ denote the free filtered commutative R-algebra with an element $z^{1\langle 1\rangle}$ in double degree $(1,1)$. Then, for any $\alpha \in \pi_1 R$, the image of α in filtration degree 1 of $\mathrm{ct}(R)$ determines a map of filtered objects. We define $R^\alpha_\bullet$ as the pushout*

$$\begin{array}{ccc} \mathrm{ct}(R)\{z^{1\langle 1\rangle}\} & \xrightarrow{\alpha} & \mathrm{ct}(R) \\ {\scriptstyle 0}\downarrow & \ulcorner & \vdots \\ \mathrm{ct}(R) & \dashrightarrow & R^\alpha_\bullet \end{array}$$

Then the underlying object of $R^\alpha_\bullet$ is $R//\alpha$ and the fiber of $\mathrm{ct}(R) \to R^\alpha_\bullet$ is purely odd.

Proof. The first statement follows immediately because we can apply colim to the defining diagram. For the purely odd part, note that the graded pieces of $\mathrm{ct}(R)\{z^{1\langle 1\rangle}\}$ is computed by

$$\mathrm{gr}(\mathrm{ct}(R)\{z^{1\langle 1\rangle}\})_i \cong (\Sigma R)^{\otimes i}_{h\Sigma_i}$$

Applying the graded pieces functor to the defining diagram, we obtain

$$\begin{array}{ccc} \mathrm{gr}(\mathrm{ct}(R)\{z^{1\langle 1\rangle}\}) & \xrightarrow{0} & \mathrm{gr}(\mathrm{ct}(R)) \\ {\scriptstyle 0}\downarrow & \ulcorner & \downarrow \\ \mathrm{gr}(\mathrm{ct}(R)) & \longrightarrow & \mathrm{gr}(R^\alpha_\bullet) \end{array}$$

By which we compute

$$\mathrm{gr}(R^\alpha_\bullet)_i \cong (\Sigma^2 R)^{\otimes i}_{h\Sigma_i}$$

The purely odd part then follows by checking the rank of graded pieces. □

3. Proof of Theorem A

We now prove the Theorem A of [BSY22]. We recall the definition of Nullstellensatzian object in a general category.

Definition 3.1. We say that a presentable ∞-category $\mathcal{C}$ is **Nullstellensatzian** if every compact and non-terminal object admits a map to the initial object. We say that an object R in a presentable ∞-category $\mathcal{C}$ is Nullstellensatzian if R is non-terminal and $\mathcal{C}_{R/}$ is Nullstellensatzian.

The classical Hilbert Nullstellensatz states that the Nullstellensatzian objects in CRing are exactly algebraic closed fields. We will prove the following:

Theorem 3.2. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ be a nonzero object. R is Nullstellensatzian if and only if $R = E(L)$ for some algebraic closed field L.*

Bofore going on, we explain a property in which the Nullstellensatzian object in a presentable category mimicing algebraic closed fields.

Proposition 3.3. *Let $\mathcal{C}$ be a presentably symmetric monoidal ∞-category and let $R \in \mathrm{CAlg}(\mathcal{C})$ be Nullstellensatzian. Suppose W_1, W_2 are compact R-modules and $T \in \mathrm{CAlg}_R(\mathcal{C})$, we let $0 : R\{W\} \to T$ denote the R-algebra map induced the zero map $W \to T$. Then, for any algebra map $P\colon R\{W_1\} \to R\{W_2\}$, the following are equivalent:*

(1) There exists a nonzero $T \in \mathrm{CAlg}_R(\mathcal{C})$ and a factorization in $\mathrm{CAlg}_R(\mathcal{C})$:

$$\begin{array}{ccc} R\{W_1\} & \xrightarrow{P} & R\{W_2\} \\ & \searrow^{0} & \downarrow \\ & & T \end{array}$$

(2) There exists a factorization in $\mathrm{CAlg}_R(\mathcal{C})$:

$$\begin{array}{ccc} R\{W_1\} & \xrightarrow{P} & R\{W_2\} \\ & \searrow^{0} & \downarrow \\ & & R \end{array}$$

Proof. It suffices to show that (1) implies (2). Since W_1, W_2 are compact, it follows that the algebras $R\{W_1\}$ and $R\{W_2\}$ are also compact in $\mathrm{CAlg}_R(\mathcal{C})$. By assumption, we have a factorization

$$\begin{array}{ccc} R\{W_1\} & \xrightarrow{P} & R\{W_2\} \\ {\scriptstyle 0}\downarrow & & \downarrow \\ R & \longrightarrow & T \end{array}$$

Denote

$$Q := R \otimes_{R\{W_1\}} R\{W_2\}$$

Then Q is compact in $\mathrm{CAlg}_R(\mathcal{C})$ and $Q \neq 0$. Thus, the condition that R is Nullstellensatzian inmplies there exists a retract $R \to Q \to R$ which gives the desired factorization. □

One can think of P as an equation. The above proposition amounts to say that if R is Nullstellensatzian, then an equation has a solution in R if and only if it has a solution in some of its extension, which is exactly a defining property of an algebraic closed field.

Corollary 3.4. *Fix the notations as above, then the induced map*

$$\mathrm{map}_{\mathcal{C}}(W, R) \to \mathrm{map}_{\mathcal{C}}(W, T)$$

on mapping spectra is injective on homotopy groups.

Proof. Suppose $x \in \ker(\pi_k \mathrm{map}_{\mathcal{C}}(W, R) \to \pi_k \mathrm{map}_{\mathcal{C}}(W, T))$. x defines a map of algebra

$$x : R\{\Sigma^k W\} \to R$$

such that the composite

$$R\{\Sigma^k W\} \xrightarrow{x} R \to T$$

is homotopic to 0. By the proposition, $x : R\{\Sigma^k W\} \to R$ is homotopic to $0 : R\{\Sigma^k W\} \to R$. □

Remark 3.5. Suppose we additionally require $\mathcal{C}$ to be rigid, then the above corollary has a dual form, that means, the map

$$W \otimes R \to W \otimes E(L)$$

is injective on homotopy groups.

Now we can prove Theorem 3.2. Since we have a description of the essential image of the Lubin-Tate functor, it suffices to check that any Nullstellensatzian object in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ satisfies these properties. To begin with, we have:

Lemma 3.6. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ be Nullstellensatzian. Then R is even.*

Proof. By Corollary 2.2, there exists a map $R \to E(L)$ for some algebraic closed field L. Choose V_n to be the generalized Smith-Toda complex. By Corollary 3.4 and Remark 3.5, the map

$$V_n \otimes R \to V_n \otimes E(L)$$

is injective on homotopy groups. Therefore $V_n \otimes R$ is even, which implies R is even. □

As a consequence, on can choose $v_0, \cdots, v_{n-1} \in \pi_* R$ and define

$$K := R/v_0 \otimes_R \cdots \otimes_R R/v_{n-1}$$

Lemma 3.7. *K is a compact R module.*

Proof. K is generated under finite colimit of R, so it is dualizable, which implied $K \otimes V_n$ is compact. Since K is a retract of $K \otimes V_n$, K is compact. □

Now it remains to show the following.

Proposition 3.8. *Let $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ be Nullstellensatzian. Then the graded ring $\pi_* K$ is even periodic and $\pi_0(K)$ is an algebraically closed field.*

Proof. We first show that $\pi_* K$ is even and commutative. Choose a map $R \to E(L)$ for some algebraically closed field L. Since K is compact, the extension

$$\iota : K \to K \otimes_R E(L) \simeq E(L)//\mathfrak{m}$$

is injective on homotopy groups. This implies that $\pi_* K$ is even commutative.

To show the remaining parts of the proposition, we show that any even graded polynomial $f \in \pi_*(K)[t_1, \cdots, t_l]$ has a solution in $\pi_* K$. One can regard f as a natural transformation

$$f : \prod_{i=1}^{l} \pi_{2d_i}(- \otimes_R K) \to \pi_{2d}(- \otimes_R K)$$

from $\mathrm{CAlg}_R^{\wedge}$ to $\mathrm{Fun}(\mathbb{Z}, \mathrm{Set})$, where we fix $\Sigma d_i = d$. By Yoneda, f is represented by

$$P_f : R\{\Sigma^{2d} K^{\vee}\} \to R\{\bigoplus_{i=1}^{l} \Sigma^{2d_i} K^{\vee}\}$$

The condition that f has a solution is equivalent to say that we have a factorization

$$\begin{array}{ccc} R\{\Sigma^{2d}K^\vee\} & \xrightarrow{P_f} & R\{\bigoplus_{i=1}^{l}\Sigma^{2d_i}K^\vee\} \\ & \searrow^{0} & \downarrow \\ & & \pi_*(K) \end{array}$$

Note that through the extension $K \to K \otimes_R E(L) \simeq E(L)//\mathfrak{m}$, f always has a solution in $\pi_*(E(L)//\mathfrak{m}) \cong L[u^{\pm 1}]$ since L is algebraic closed. It follows from Proposition 3.3 that f has a solution in $\pi_*(K)$.

Therefore, if we take $f = at - 1$, that f has a solution means $\pi_0(K)$ is a field. Taking $f = t_1t_2 - 1$ with t_1 in degree 2 shows that π_*K is periodic, and taking f to be any ungraded polynomial shows that π_0K is algebraic closed. □

Combining the above consequences and the description of the essential image of $E(-)$, we proved Theorem 3.2.

Remark 3.9. One can define Nullstellensatzian for general regular cardinalities. The similar statement of Theorem 3.2 holds true for any regular cardinality.

4. Constructable spectra

We briefly introduce the constructable spectra theory.

The Theorem D shows that there are enough geometric points in chromatic world, and the Theorem A implies that all the geometric points are closed. It brings a natural way for us to study the geometry of these chromatic rings. The first step is to attempt to define a topological space $\mathrm{Spec}(R)$ for $R \in \mathrm{CAlg}(\mathrm{Sp}_{T(n)})$.

We introduce the following definition.

Definition 4.1. Let $\mathcal{C}$ be the category of products of algebraically closed fields. These fields are equipped with a formal group of height n, which is unique up to isomorphism. We define the **constructible spectrum** functor

$$\mathrm{Spec}_{T(n)}(-) : \mathrm{CAlg}(\mathrm{Sp}_{T(n)})^{\mathrm{op}} \to \mathrm{CHaus}$$

as the left Kan extension:

$$\begin{array}{ccc} \mathcal{C}^{\mathrm{op}} & \xrightarrow{\mathrm{Spec}(-)} & \mathrm{CHaus} \\ & \searrow^{E(-;-)} \quad \nearrow_{\mathrm{Spec}_{T(n)}(-)} & \\ & \mathrm{CAlg}(\mathrm{Sp}_{T(n)})^{\mathrm{op}} & \end{array}$$

Theorem 4.2. *The constructible spectrum functor* $\mathrm{Spec}_{T(n)}(-)$ *enjoys the following properties:*

1. $\mathrm{Spec}_{T(n)}(R)$ *is empty if and only if* $R = 0$.
2. *If* L *is an algebraically closed field, then* $\mathrm{Spec}_{T(n)}(E(L))$ *is a point.*
3. *For every point* $q \in \mathrm{Spec}_{T(n)}(R)$*, there exists an algebraically closed field* L *and a map* $R \to E(L)$ *such that the image of the map*

$$\{*\} \cong \mathrm{Spec}_{T(n)}(E(L)) \to \mathrm{Spec}_{T(n)}(R)$$

is $\{q\}$.

(4) *A subset* $U \subset \mathrm{Spec}_{T(n)}(R)$ *is closed if and only if there exists a map* $R \to S$ *such that* U *is the image of the map*

$$\mathrm{Spec}_{T(n)}(S) \to \mathrm{Spec}_{T(n)}(R)$$

(5) *Given a span* $S \leftarrow R \to T$*, the natural comparison map*

$$\mathrm{Spec}_{T(n)}\left(S \otimes_R T\right) \to \mathrm{Spec}_{T(n)}(S) \times_{\mathrm{Spec}_{T(n)}(R)} \mathrm{Spec}_{T(n)}(T)$$

is surjective.

These reflect basic properties of the classical Zariski spectrum.

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